MindCare

Project Report

Text-Based Mental Well-Being Analysis Using Deep Learning

Benchmarks: Multi-Class Emotion (6 classes) and Reddit Mental Health (Distress vs. Control)

Project Title: MindCare — Text-Based Mental Well-Being Analysis Using Deep Learning

Task: Emotion classification and mental-distress detection from text

Datasets: `dair-ai/emotion` (6 classes) and `kamruzzaman-asif/reddit-mental-health-classification` (binary)

Model: DistilBERT + BiGRU + Additive Attention

Stack: Python, PyTorch, Hugging Face Transformers, Streamlit, SQLite, Plotly, ReportLab

Code: `https://github.com/Visshu78/MindCare/`

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1 Project Overview

Project Objective

This project investigates whether a lightweight hybrid deep learning model can identify emotional distress in everyday written text while remaining fast enough for real-time use and transparent enough to explain its predictions. The system, **MindCare**, combines a pretrained transformer (DistilBERT [4]), a bidirectional recurrent layer (BiGRU [6]), and an additive attention layer [5], and is deployed inside an end-to-end journaling and analytics web application.

The work compares four architectures under identical training conditions on two benchmarks:

1. **Multi-Class Emotion benchmark** — six emotions: sadness, joy, love, anger, fear, surprise.
2. **Reddit Mental Health benchmark** — binary detection of mental-distress posts versus control posts.

Background and Motivation

Stress, anxiety, loneliness and emotional distress often show up in the way people write, but the signals are subtle and context-dependent. Keyword approaches miss polysemy, sarcasm and long-range context, while standalone transformers use only a pooled `[CLS]` vector that hides which words drove a decision. MindCare targets this gap by adding sequential modelling and token-level attention on top of a compact transformer. Related work on transformer-based and hybrid architectures for mental-disorder and emotion detection includes [3, 8, 7], and DistilBERT-based emotion classifiers have been studied in [9, 10].

The guiding research questions were:

- Can everyday text be used to identify nuanced emotional distress?
- Can a lightweight hybrid model keep transformer-grade understanding with low latency and token-level interpretability?

> **Scope note.** MindCare is an early-awareness and self-reflection tool. It is *not* a clinical diagnostic system and does not replace professional assessment.

Code Availability

The complete source code, including the training scripts, the Streamlit application and the experiment results, is available at `https://github.com/Visshu78/MindCare/`.

2 System Architecture

The system has two parts: the **research core** (model training and evaluation, `model_experiments/`) and the **web application** (`MindCare_v2/`).

2.1 Model Architecture Overview

MindCare uses a hierarchical hybrid classifier in which each component adds a distinct capability to the previous one. Throughout, B is the batch size, L the maximum sequence length (48 for emotion, 128 for Reddit) and C the number of classes (6 for emotion, 2 for Reddit). Figure 1 shows the proposed architecture with the tensor shape at every stage.

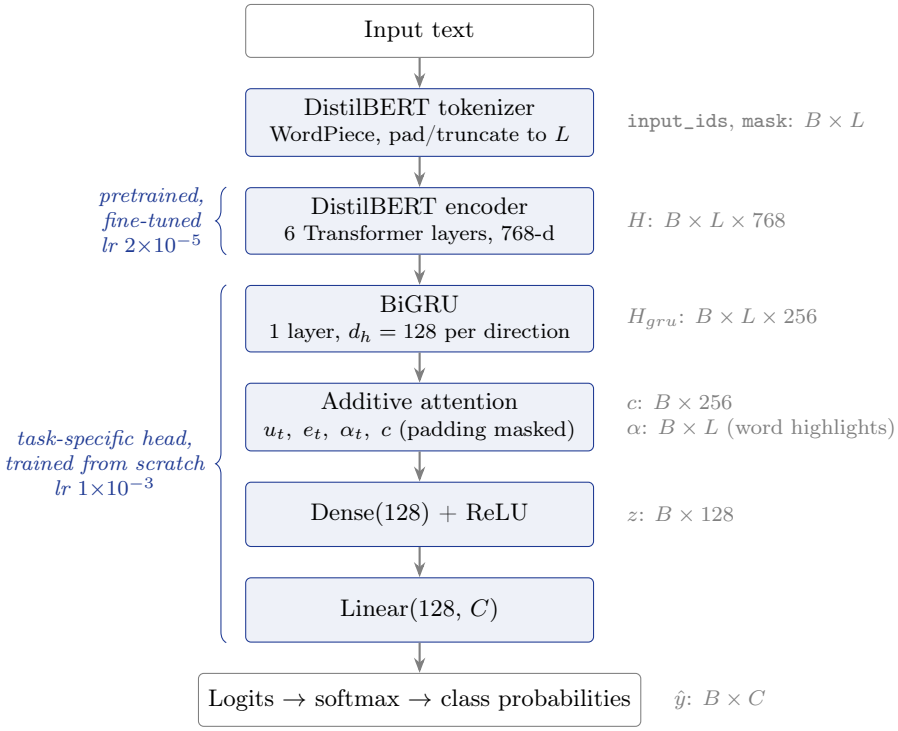


Figure 1: Proposed MindCare architecture (DistilBERT + BiGRU + Additive Attention) with tensor shapes.

Stage-by-stage description.

1. **Tokenisation.** The DistilBERT fast tokenizer converts the text to sub-word token IDs with an attention mask, padded or truncated to L tokens.
2. **Contextual encoding.** The pretrained `distilbert-base-uncased` encoder (6 Transformer layers, 768-d hidden states) produces one context-aware vector per token, $H \in \mathbb{R}^{B\times L\times 768}$. The backbone is fine-tuned end-to-end at a low learning rate.
3. **Sequential modelling.** A single-layer bidirectional GRU ($d_h = 128$ per direction) reads the token vectors left-to-right and right-to-left, so each position carries both preceding and following context: $H_{gru} \in \mathbb{R}^{B\times L\times 256}$. This captures how emotion builds or shifts across clauses.
4. **Attention pooling.** Additive attention scores every token, masks padding, normalises the scores with a softmax and returns the weighted sum $c \in \mathbb{R}^{B\times 256}$. The weights α show which words drove the prediction (Section 8).
5. **Classification head.** Dense(128) with ReLU followed by Linear(128, C) produces the logits; a softmax converts them to class probabilities. Training uses cross-entropy loss.

2.2 Layer-wise Specification

Component parameter counts below are computed from the layer dimensions and sum to the reported total of 67,119,238 for the six-class model.

Stage	Operation	Output shape	Parameters
Encoder	DistilBERT, 6 layers	$B \times L \times 768$	66,362,880
Sequence layer	BiGRU, $d_h = 128$	$B \times L \times 256$	689,664
Attention	$W_a \in \mathbb{R}^{128\times256}$, $v_a \in \mathbb{R}^{128}$	$B \times 256$	33,024
Hidden layer	Linear(256, 128) + ReLU	$B \times 128$	32,896
Output layer	Linear(128, C=6)	$B \times 6$	774
Total			**67,119,238**

2.3 Design Rationale

- **DistilBERT** keeps most of BERT's language understanding with about 40% fewer parameters and roughly 60% faster inference, which suits real-time journaling feedback.
- **BiGRU** adds explicit sequential, two-directional modelling on top of the transformer features, which a single `[CLS]` vector does not provide.
- **Additive attention** replaces uniform or `[CLS]` pooling with learned token weights, so emotionally loaded words (for example *hopeless* or *terrified*) contribute more, and the weights can be displayed to the user.
- **A small dense head with a higher learning rate** lets the new layers adapt quickly while the pretrained backbone changes slowly.

2.4 Compared Architectures

Model	Description	Params (C=6)
DistilBERT Baseline	`[CLS]` hidden state → Dropout(0.2) → Linear(768, C)	66.37M
DistilBERT + BiGRU	All token states into a 1-layer BiGRU (128); final forward and backward states concatenated (256-d) → Dropout → Linear	67.05M
Proposed: + Attention	BiGRU outputs pooled by additive attention (256-d) → Dense(128)+ReLU → Linear	67.12M
+ CNN + BiGRU + Attn	1D convolution (kernel 3, 256 channels) before the BiGRU	67.32M

2.5 Additive Attention

The attention layer follows the additive formulation of Bahdanau et al. [5]. For each token t with BiGRU output $h_t \in \mathbb{R}^{256}$:

$$u_t = \tanh(W_a h_t + b_a), \qquad e_t = v_a^\top u_t, \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{L} \exp(e_k)}, \qquad c = \sum_{t=1}^{L} \alpha_t h_t.$$

Padding positions are masked ($e_{pad} = -10^9$) before the softmax. The weights α_t are reused directly for explainability (Section 8).

2.6 Application Architecture

The Streamlit application (`main.py`) routes between five tabs:

- **Journal** — daily entry, 1–10 mood slider, real-time prediction and empathetic feedback.
- **History** — past entries with emotion filtering and PDF export (ReportLab).
- **Analytics** — Plotly mood-over-time chart, emotion distribution, journaling streak.
- **Evaluation** — ablation matrix, confusion-matrix heatmaps and a live attention-inspector.
- **Settings** — profile and account security.

User data is stored in SQLite with foreign-key constraints, and passwords are hashed with PBKDF2-HMAC and a salt.

3 Datasets

3.1 Benchmark 1 — Multi-Class Emotion

Source: `dair-ai/emotion` (Saravia et al. [2]), English Twitter statements labelled with six emotions. A lightweight stratified split was used:

Split	Samples	Notes
Train	2,500	stratified over 6 classes
Validation	500	stratified
Test (held-out)	500	stratified, unseen in training

Tokenisation uses the DistilBERT fast tokenizer (`distilbert-base-uncased`) with maximum length 48.

3.2 Benchmark 2 — Reddit Mental Health

Source: `kamruzzaman-asif/reddit-mental-health-classification`, following the methodology of Hasan et al. [1] (see also [11] for analysis of Reddit mental-health communities). The task is binary: **Mental Distress** (clinical subreddits such as depression, health anxiety, bipolar, BPD, addiction, alcoholism) versus **Control** (non-clinical subreddits such as personal finance, jokes, teaching, meditation, relationships).

Preprocessing removed URLs, markdown links, blockquote artefacts and extra whitespace, and dropped posts under 30 characters or 5 words.

Split	Samples	Balance
Train	3,000	1,500 / 1,500
Validation	600	300 / 300
Test (held-out)	600	300 / 300

Maximum sequence length is 128 tokens.

4 Experimental Design

All four architectures were trained with one shared harness so that the architecture is the main experimental variable.

Setting	Value
Backbone	`distilbert-base-uncased`
Max sequence length	48 (emotion), 128 (Reddit)
Batch size / epochs	32 / 3
Optimizer	AdamW, weight decay 0.01
Learning rates	2×10^{-5} backbone, 1×10^{-3} head (differential)
Gradient clipping	max norm 1.0
Loss	Cross-entropy
Checkpoint selection	Best validation macro F1

4.1 Evaluation Protocol

- Metrics: accuracy, macro F1, weighted F1, macro precision, macro recall, per-sample latency, parameter count.
- Confusion matrices and class-wise precision/recall/F1 for the proposed model.
- Differences between percentage-valued metrics are reported in percentage points (pts), not relative change.

5 Training Results

Validation results at the final epoch (epoch 3) are shown below; every model improved steadily across the three epochs.

5.1 Multi-Class Emotion

Model	Train acc.	Val acc.	Val macro F1	Time (s)
DistilBERT Baseline	91.64%	88.40%	84.62%	502.4
DistilBERT + BiGRU	92.68%	89.20%	85.49%	562.4
Proposed (+ Attention)	90.88%	89.20%	**87.05%**	562.0
+ CNN + BiGRU + Attn	89.24%	86.40%	82.34%	558.7

5.2 Reddit Mental Health

Model	Train acc.	Val acc.	Val macro F1	Time (s)
DistilBERT Baseline	98.37%	94.00%	94.00%	4,264.3
DistilBERT + BiGRU	98.57%	94.00%	94.00%	1,494.0
Proposed (+ Attention)	98.47%	94.17%	94.16%	1,283.7

On Reddit the baseline's validation loss rose from 0.1980 to 0.2840 while its training loss kept falling, a sign of mild overfitting by epoch 3. Training times differ widely across runs and should not be treated as a clean speed comparison.

6 Test Results

6.1 Multi-Class Emotion (500 held-out samples)

Model	Acc.	Macro F1	Wtd F1	Macro P	Macro R	Latency
DistilBERT Baseline	89.40	84.12	89.24	86.29	82.39	17.02 ms
DistilBERT + BiGRU	**89.80**	84.37	**89.61**	87.83	83.13	18.35 ms
Proposed	89.40	**85.62**	89.44	87.25	**85.28**	18.45 ms
+ CNN + BiGRU + Attn	89.60	84.73	89.35	**88.62**	82.51	18.41 ms

6.2 Reddit Mental Health (600 held-out samples)

Model	Acc.	Macro F1	Macro P	Macro R	Test loss	Latency
DistilBERT Baseline	95.00	95.00	95.00	95.00	**0.1358**	38.99 ms
DistilBERT + BiGRU	**95.50**	**95.50**	**95.54**	**95.50**	0.1783	**36.65 ms**
Proposed	94.00	94.00	94.05	94.00	0.2442	37.12 ms

6.3 Class-Wise Results (Proposed Model, Emotion)

Class	Precision	Recall	F1	Support
Sadness	91.84	93.10	92.47	145
Joy	93.45	90.23	91.81	174
Love	68.75	84.62	75.86	39
Anger	93.44	82.61	87.69	69
Fear	84.38	96.43	90.00	56
Surprise	91.67	64.71	75.86	17
Macro avg	87.25	85.28	85.62	500
Weighted avg	89.84	89.40	89.44	500

6.4 Confusion Matrices

Emotion — proposed model (rows: true, columns: predicted)

	Sad	Joy	Love	Anger	Fear	Surprise
Sadness	**135**	4	1	3	2	0
Joy	3	**157**	14	0	0	0
Love	1	4	**33**	1	0	0
Anger	6	1	0	**57**	5	0
Fear	1	0	0	0	**54**	1
Surprise	1	2	0	0	3	**11**

Reddit — all models (rows: true; columns: predicted Control / Distress)

Model	True Control (300)	True Distress (300)
DistilBERT Baseline	286 / 14	16 / 284
DistilBERT + BiGRU	282 / 18	9 / 291
Proposed	287 / 13	23 / 277

7 Comparative Analysis

Multi-Class Emotion

Relative to the DistilBERT baseline, the proposed model had the same test accuracy (89.40%) but higher macro F1 (+1.50 pts), macro recall (+2.89 pts), macro precision (+0.96 pts) and weighted F1 (+0.20 pts), at a cost of 1.43 ms extra latency per sample. The BiGRU-only model recorded the highest accuracy and weighted F1, so no single variant led on every metric. The gains concentrate in macro-averaged metrics, which weight small classes equally, so they are consistent with better handling of minority classes; the 500-sample test set (only 17 *surprise* examples) is small enough that this should be read as an indication rather than a proof.

The CNN variant added parameters without beating BiGRU + Attention on macro F1 (84.73% vs. 85.62%), so the extra convolution did not pay off in this setup.

Reddit Mental Health

The ordering reverses here. DistilBERT + BiGRU led on accuracy and all macro metrics (+0.50 pts over the baseline), while the proposed attention model was 1.00 pts below the baseline on accuracy and F1 and had the highest test loss. In the confusion matrices, the proposed model missed more distress posts (23 false negatives vs. 16 for the baseline and 9 for BiGRU).

Interpretation

The benefit of attention is therefore *dataset-dependent*: it helped macro-level balance on short emotional statements and did not help on long-form Reddit posts. It should be described as an observed difference, not as evidence that one architecture is universally better. Attention's most consistent contribution across both benchmarks is interpretability rather than raw accuracy.

8 Explainability

Because attention weights are normalised over tokens, the model can display which words contributed most to a prediction. Examples from the application:

Input	Prediction	Top attended tokens
"I feel completely overwhelmed and hopeless…"	Sadness (98.4%)	hopeless, overwhelmed, feel
"I am really terrified about failing my upcoming test…"	Fear (99.1%)	terrified, failing, test
"Had a wonderful time celebrating with close friends…"	Joy (97.8%)	wonderful, celebrating, calm

These highlights are shown in the journal feedback card and in the live tester on the Evaluation tab. Attention weights indicate where the model looked, which is useful for review but is not a guarantee of causal importance.

9 Safety and Crisis Intervention

A rule-based interceptor runs on every entry before normal feedback. If the text contains crisis terms (e.g. "suicide", "kill myself", "want to die", "end it all"), the entry is tagged `CRISIS`, standard feedback is suppressed, and an emergency banner with helpline guidance is shown instead. This layer is deliberately independent of the neural model so that it cannot be bypassed by a low-confidence prediction.

10 Key Findings

1. **Strong results at low cost.** All models reached about 89–90% accuracy on six emotions and 94–95.5% on Reddit distress detection, with 17–39 ms latency per sample.
2. **Best macro F1 on emotions.** The proposed model achieved 85.62% macro F1 and 85.28% macro recall, the highest of the four.

3. **Fear sensitivity.** Recall on *fear* was 96.43% and on *sadness* 93.10%, the two highest per-class recalls.
4. **Low overhead.** BiGRU plus attention added about 0.7M parameters and 1.43 ms latency over the baseline.
5. **Mixed ablation outcome.** BiGRU alone was best on Reddit and on emotion accuracy; attention was best on emotion macro metrics.
6. **CNN did not help.** The extra convolution layer reduced macro F1 relative to the proposed model.
7. **Weakest classes.** *Love* (F1 75.86%) and *surprise* (F1 75.86%, recall 64.71%) remain the hardest, partly because of small support.

11 Final Results at a Glance

Multi-Class Emotion — Proposed vs. DistilBERT Baseline

Metric	Baseline	Proposed	Difference
Accuracy	89.40%	89.40%	0.00 pts
Macro F1	84.12%	85.62%	+1.50 pts
Weighted F1	89.24%	89.44%	+0.20 pts
Macro precision	86.29%	87.25%	+0.96 pts
Macro recall	82.39%	85.28%	+2.89 pts
Latency	17.02 ms	18.45 ms	+1.43 ms

Reddit Mental Health — Proposed vs. DistilBERT Baseline

Metric	Baseline	Proposed	Difference
Accuracy	95.00%	94.00%	−1.00 pts
Macro F1	95.00%	94.00%	−1.00 pts
Macro precision	95.00%	94.05%	−0.95 pts
Macro recall	95.00%	94.00%	−1.00 pts
Latency	38.99 ms	37.12 ms	−1.87 ms

12 Conclusion

MindCare demonstrates a complete workflow from dataset curation through controlled model ablation, explainability and deployment in a usable application. The proposed DistilBERT + BiGRU + Additive Attention model reached 85.62% macro F1 on six-class emotion recognition with only a small latency cost and provides token-level explanations, while on the Reddit distress task the simpler BiGRU variant performed best. The main conclusion is that the value of attention depended on the dataset: it improved class balance on short emotional text, and its consistent benefit was transparency.

Future work: multimodal input (voice journaling with HuBERT/Wav2Vec2), longitudinal modelling of month-scale mood trajectories, INT8 quantisation via ONNX/TensorRT for on-device

inference, federated learning for privacy-preserving improvement, training on the full datasets with multiple seeds and significance testing, and a learned crisis classifier alongside the rule-based layer.

13 Limitations and Interpretation

1. **Small training subsets.** Training used 2,500 (emotion) and 3,000 (Reddit) samples, only a fraction of the available corpora, and only 3 epochs.
2. **Single run, no significance testing.** Results are point estimates from one run on 500- and 600-sample test sets; differences of about 1 pt are within plausible noise.
3. **Text only.** No voice, facial or physiological signals are used.
4. **Short context.** Token limits (48–128) suit single entries but not long-term histories.
5. **Keyword crisis rule.** Substring matching is simple and transparent but can misfire (for example on the word "die" in unrelated contexts) and can miss indirect expressions of crisis.
6. **Proxy labels.** Reddit labels come from subreddit membership, not clinical diagnosis, and emotion labels come from Twitter annotations; neither establishes a person's mental health state.
7. **Not a clinical tool.** Outputs are for self-reflection and should not replace professional care.

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